9th Class • Punjab Board • PCTB New Syllabus

Chapter 5: Tissues, Organs, and Organ Systems

Solved notes in book order — MCQs, short answers, detailed answers, inquisitive questions and key points.

Chapter 5 • Units 5.1 – 5.4 • Exercise solved in book order

What this chapter is about

• In a multicellular organism, cells are not independent. They are organized at different levels: cells, tissues, organs, organ systems and, finally, the organism.

• The chapter explains the four types of animal tissues (epithelial, connective, muscular, nervous) and the plant tissues (epidermal, ground and vascular), and then the tissues inside the stomach and the leaf as examples.

• It also describes the organs and organ systems of plants (root and shoot system) and of humans (11 systems), and finally homeostasis — how organ systems work together to keep the internal environment stable.

• The textbook exercise (10 MCQs, 8 short answers with four differences, 10 detailed answers and 2 inquisitive questions) is solved here in book order.

A. Multiple Choice Questions

Tick the correct option. The reason after each answer tells you why the other options are wrong.

1. A higher level of organization exhibits emergent properties when:

a) Its parts function independently

b) The sum of its parts is greater than the whole

c) The individual parts are more important than the whole

d) Its parts interact to perform more functions

Answer: d) Its parts interact to perform more functions — Emergent properties appear only when the parts work together and interact; the new property is not present in any single part alone.

2. Which of the following demonstrates the levels of organization of the body, from simplest to most complex?

a) Cell → Organ → Tissue → Organelle → Organ system

b) Organelle → Cell → Tissue → Organ → Organ system

c) Tissue → Cell → Organelle → Organ → Organ system

d) Organ system → Tissue → Cell → Organelle → Organ

Answer: b) Organelle → Cell → Tissue → Organ → Organ system — The correct order from simplest to most complex is organelle, cell, tissue, organ, organ system, and then organism.

3. At which level of organization gas exchange occurs between body and environment?

a) Organelle level in mitochondria

b) Cellular level in alveolar cells

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c) Tissue level in epithelial tissues
d) Organ system level in the respiratory system

Answer: d) Organ system level in the respiratory system — Gas exchange between the body and the environment is a function of the respiratory system, which is an organ system made of lungs, trachea and associated organs.

4. The epithelial tissue in the stomach wall is responsible for producing:

a) Mucus
b) Pepsinogen
c) Hydrochloric acid
d) All of these

Answer: d) All of these — The epithelial cells of the stomach secrete mucus (protection), pepsinogen (an enzyme) and hydrochloric acid (for the acidic medium).

5. In the wall of stomach, which tissue also contains blood vessels and nerves?

a) Epithelial
b) Muscle
c) Inner connective
d) Outer connective

Answer: c) Inner connective — The inner connective tissue layer of the stomach wall carries blood vessels and nerves which supply the working tissues.

6. In a leaf, which tissue is responsible for photosynthesis?

a) Xylem
b) Mesophyll
c) Epidermis
d) Phloem

Answer: b) Mesophyll — The mesophyll (palisade and spongy) is the ground tissue of the leaf; its cells contain chloroplasts and carry out photosynthesis.

7. What is the primary function of the xylem tissue in a leaf?

a) To transport sugars to other parts
b) To transport water to parts of leaf
c) To synthesize chlorophyll
d) To control the opening and closing of stomata

Answer: b) To transport water to parts of leaf — Xylem carries water and mineral salts from the roots to the leaves; sugars are carried by phloem and stomata are controlled by guard cells.

8. Which of these is a function of the human skeletal system?

a) Storing minerals and producing blood cells
b) Removing carbon dioxide from blood
c) Filtering blood to remove waste products
d) Breaking down food for energy

Answer: a) Storing minerals and producing blood cells — Bones store minerals such as calcium and phosphate, and the bone marrow produces blood cells; the other three jobs belong to the respiratory, excretory and digestive systems.

9. Which structures are responsible for the transport of food in plant body?

a) Xylem tissue
b) Palisade mesophyll
c) Phloem tissue
d) Spongy mesophyll

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Answer: c) Phloem tissue — Phloem is the vascular tissue that transports the food (glucose made in the leaves) to all parts of the plant.

10. In a plant, which of the following is the primary function of the flower?

a) Transporting water and minerals
b) Supporting leaf growth
c) Facilitating reproduction through pollination
d) Regulating gas exchange

Answer: c) Facilitating reproduction through pollination — The flower is the reproductive organ of the plant; pollination and fertilization take place in it, and it forms the fruit and seed.

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B. Write Short Answers

1. Enlist the levels of organization from cells to organ systems.

The levels are: **organelle → cell → tissue → organ → organ system → organism**. In detail: (1) Molecules and organelles work together inside the cell; (2) cells are the basic units of structure and function; (3) a tissue is a group of similar cells that work together to perform a specific function; (4) an organ is a structure made of tissues that work together to perform specific functions; and (5) an organ system is a group of organs that work together to perform a particular function, e.g. the digestive system.

2. What are the major roles of the epithelial tissue present in the stomach?

1. Protection: it lines the inner surface of the stomach and protects the underlying tissues from the strong hydrochloric acid and from mechanical injury.

2. Secretion: its cells secrete mucus, hydrochloric acid and the enzyme pepsinogen, which are needed for the digestion of proteins.

3. Absorption: it allows the absorption of water, salts and some digested food.

4. Lubrication: the mucus it secretes keeps the surface slippery so that food can move easily and the wall is not digested by its own enzymes.

3. How do the smooth muscles contribute to the stomach's function?

Smooth muscle tissue is present in the wall of the stomach in three layers (inner circular, middle oblique and outer longitudinal). By their contraction and relaxation they:

1. produce the churning movements (peristalsis) which mix the food with gastric juice and break it into a semi-liquid mass (chyme);

2. push the food forward into the small intestine; and

3. help in the mechanical digestion of food. These muscles are involuntary, so this work goes on without our will.

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4. What is the function of the palisade mesophyll in the leaf?

Palisade mesophyll cells are elongated, tightly packed cells present just below the upper epidermis. They contain a large number of chloroplasts and are the main site of photosynthesis. Because they lie near the upper surface, they receive the maximum light, and their shape and arrangement allow maximum absorption of light energy to prepare glucose for the plant.

5. What is the role of the shoot system in plants?

The shoot system is the part of the plant above the ground. It consists of the stem, leaves, flowers and fruits. Its roles are:

1. the stem supports the leaves, flowers and fruits and keeps them in the air for light;

2. leaves prepare food by photosynthesis and exchange gases through stomata;

3. it transports water, minerals and prepared food between the roots and the leaves (through xylem and phloem); and
4. flowers take part in reproduction — they are the reproductive organs which form fruits and seeds.

6. What is homeostasis, and why is it important for organisms?

Definition: homeostasis is the ability of an organism (or a cell) to maintain a relatively stable internal environment despite changes in the external environment. It keeps conditions such as body temperature, pH and the concentration of water and solutes within a narrow, suitable range.
Importance: (i) the enzymes and chemical reactions of the body work only in a particular range of temperature and pH; (ii) cells need the right amount of water, salts and glucose to stay alive; (iii) it protects the body from the harmful effects of heat, cold, water loss and wastes; and (iv) it allows the organism to live in changing surroundings. Without homeostasis, the body's metabolism would stop and the organism would die.

7. How does the human body maintain a stable internal temperature?

The normal body temperature of a human is about 37 °C, and it is kept stable by the temperature regulating centre present in the brain (hypothalamus). It works like a thermostat:
1. When the body temperature falls (in cold weather), the blood becomes cool. The temperature regulating centre sends nerve impulses to the muscles, which start to shiver, and the muscles produce heat. The blood vessels of the skin also become narrow (vasoconstriction) so that less heat is lost from the surface.
2. When the body temperature rises (in hot weather or after exercise), the temperature regulating centre sends impulses to the sweat glands which produce sweat; the evaporation of sweat cools the skin. The blood vessels of the skin widen (vasodilation) so that more heat is lost to the surroundings.
In this way the body temperature is kept near 37 °C, which is necessary for the enzymes to work properly.

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8. Differentiate between the following: (i) Tissue and organ (ii) Root system and shoot system (iii) Epidermal and mesophyll tissue (iv) Palisade and spongy mesophyll.

Tissue	Organ
A group of similar cells that work together to perform a specific function.	A structure made of two or more types of tissues that work together to perform specific functions.
It is a simpler, lower level of organization.	It is a higher level of organization and is made of tissues.

Tissue	Organ
Examples: epithelial, connective, muscular, nervous tissue; xylem and phloem.	Examples: stomach, kidney, leaf, stem, root.

Root system	Shoot system
The part of the plant present below the ground and made mainly of roots.	The part of the plant present above the ground and made of stem, leaves, flowers and fruits.
It anchors the plant in the soil, absorbs water and mineral salts, and sometimes stores food.	It supports the plant, prepares food, transports materials and takes part in reproduction.
It is the first part that grows from the seed and grows towards gravity, away from light.	It grows upward away from gravity and towards the light.

Epidermal tissue	Mesophyll tissue
It forms the outer covering (epidermis) of leaves, stems and roots.	It is the ground tissue present between the upper and lower epidermis of the leaf.
Its cells are flat and tightly packed; guard cells and root hairs are its modified forms.	Its cells are rounded or elongated and full of chloroplasts.
It protects the inner tissues and controls the opening of stomata.	It carries out photosynthesis.

Palisade mesophyll	Spongy mesophyll
Cells are elongated (column-shaped) and tightly packed.	Cells are round and loosely arranged with many air spaces between them.
It lies just below the upper epidermis of the leaf.	It lies below the palisade mesophyll, near the lower epidermis.
It contains more chloroplasts and receives maximum light, so it carries out most of the photosynthesis.	It has fewer chloroplasts; its air spaces allow the movement of gases (carbon dioxide and oxygen).

Remember

• Animal tissues: epithelial, connective, muscular and nervous.

• Plant tissues: epidermal (protective), ground (palisade and spongy mesophyll), and vascular (xylem and phloem).

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C. Write Answers in Detail

1. Explain the levels of organization in a multicellular organism. How does each level contribute to the overall functioning of an organism?

Introduction: in a multicellular organism, cells are not independent. They are organized at different levels of complexity — cells, tissues, organs, organ systems and finally the organism. This arrangement is known as the levels of organization.

1. Organelles and molecules: within the cell, organelles such as mitochondria and ribosomes work together, and biological molecules (carbohydrates, proteins, lipids and nucleic acids) perform specific functions and work together.

2. Cell level: the cell is the basic unit of life. Individual cells do not have vital functions (processes necessary for life) except when they are in a group, so cells come together to form tissues.

3. Tissue level: a tissue is a group of similar cells that work together to perform a specific function. Each type of cell in a tissue is specialized and works as part of a team, e.g. the epithelial tissue in the stomach secretes mucus and acid.

4. Organ level: an organ is a structure made of different tissues that work together to perform specific functions. For example, the stomach has epithelial, connective, muscle and nerve tissues, and their coordination allows the stomach to hold, mix and digest food.

5. Organ system level: a group of organs that work together to perform a particular function is called an organ system. For example, the digestive system (mouth, oesophagus, stomach, small and large intestines, liver, pancreas) works together to digest and absorb food.

6. Organism level: all the organ systems work together to make up a complete (popu-lation of) organism, which can live independently.

Emergent properties: a property which appears when parts work together but which is not present in any single part is called an emergent property. In organisms, the contributions at each level do not simply add up; when the parts interact they can create new functions, capacities or achievements. Homeostasis is an example: no single cell can keep the body temperature or blood glucose constant, but the coordinated action of many organ systems achieves it.

Contribution of each level: organelles make the cell work, cells form the tissues, tissues give each organ its special functions, organs perform one job each, and organ systems coordinate different organs so that the whole organism stays alive and healthy. So each level supports the next, and the failure of one level affects all the higher levels.

2. What is a tissue level? Explain plant and animal tissues.

Tissue level: a tissue is a group of similar cells that work together to perform a specific function. At the tissue level, groups of cells have the same structure and function; for example, all muscle cells contract, so the muscle tissue produces movement.

(A) Animal tissues: there are four basic types of animal tissues.

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1. Epithelial tissue: it covers the body surface and lines the cavities and organs. Its cells are tightly packed and it protects the inner tissues, absorbs nutrients and secretes substances, e.g. the lining of the mouth, the skin and the lining of the stomach.

2. Connective tissue: it joins, supports and binds other tissues and organs. Its cells are scattered in a matrix, which may be solid (bone), flexible (cartilage) or liquid (blood). Examples: bone, cartilage, blood, adipose tissue, ligaments and tendons.

3. Muscular tissue: its cells are specialised to contract and produce movement. It is of three types: skeletal muscle (voluntary, attached to bones), cardiac muscle (involuntary, in the heart walls) and smooth muscle (involuntary, in the walls of internal organs such as the stomach and blood vessels).

4. Nervous tissue: it is made of neurons and neuroglia. It receives stimuli, conducts impulses and coordinates the activities of the body, e.g. brain, spinal cord and nerves.

(B) Plant tissues: plants have three main tissue systems.

1. Epidermal tissue: the outer protective layer of leaves, stems and roots. Its upper epidermis has a waxy cuticle to stop water loss; its lower epidermis contains guard cells which open and close the stomata for the exchange of gases and loss of water; the root epidermis has root hair cells which absorb water and minerals.

2. Ground tissue: the bulk of the leaf and stem made of mesophyll cells. In the leaf it has two parts: the palisade mesophyll (compact, many chloroplasts, main site of photosynthesis) and the spongy mesophyll (loosely arranged with air spaces, which allow the diffusion of gases).

3. Vascular tissue: the transport tissue, made of xylem and phloem. Xylem carries water and mineral salts from the roots upward and also gives support; phloem carries the prepared food (glucose) from the leaves to all parts of the plant.

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3. Describe the tissue composition of the stomach. How does each tissue contribute to the digestive function of the stomach?

The stomach is a muscular, bag-like organ that holds food and digests it. Its wall is made of four layers, i.e. four kinds of tissues, and each has its own job:

1. Epithelial tissue (inner lining): it lines the cavity of the stomach. Its cells secrete mucus, which protects the wall from the acid; hydrochloric acid, which makes the food acidic and kills bacteria; and pepsinogen, which the acid converts into pepsin to digest proteins. It also absorbs some water and salts.

2. Inner connective tissue: it lies outside the epithelium. It supports the epithelial layer, holds the blood vessels and nerves of the stomach, and so supplies food and oxygen to the working tissues and carries their messages.

3. Muscular tissue: three layers of smooth muscle (circular, oblique and longitudinal) make up the thick wall of the stomach. Their contraction churns the food with gastric juice and pushes it towards the small intestine — this is the mechanical digestion of food.

4. Outer connective tissue (serosa): the outermost layer, which covers the stomach and holds it in place, and protects it from friction with the surrounding

organs.

Conclusion: the four tissues work together: the muscle mixes the food, the epithelium digests it chemically, the inner connective tissue supplies blood and nerve messages, and the outer layer protects and supports the organ. Because of this division of labour the stomach performs its function perfectly.

4. Describe the tissue composition of the leaf. How does each tissue contribute to the functions of the leaf?

The leaf is the main food-preparing organ of the plant. Its blade is made of three tissue systems:

1. Epidermal tissue: it covers the upper and lower surfaces of the leaf. Its cells are flat, transparent and tightly packed, and a waxy cuticle covers the upper epidermis. It protects the inner tissues, reduces the loss of water, and lets the light pass through to the mesophyll. Its guard cells form the stomata, which open and close for the exchange of gases and for transpiration.

2. Ground tissue (mesophyll): it lies between the two epidermal layers and is made of two types: **palisade mesophyll** — elongated, tightly packed cells full of chloroplasts, just below the upper epidermis, which carry out most of the photosynthesis; and **spongy mesophyll** — rounded, loosely arranged cells with large air spaces, which allow carbon dioxide to reach the palisade cells and oxygen to leave them.

3. Vascular tissue: it runs through the veins (midrib and lateral veins) of the leaf. **Xylem** brings water and mineral salts from the root to the leaf for photosynthesis and also removes the extra heat; **phloem** carries the prepared food (glucose) away from the leaf to the stem, roots, flowers and fruits.

Conclusion: the epidermis protects and controls gas exchange, the mesophyll prepares the food, and the vascular tissue supplies water and carries the food away — together they make the leaf a complete food factory.

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5. How do the organ systems come together to form the human body?

The organs of the human body are grouped into organ systems. An organ system is a group of organs that work together to perform a particular function. Each system does its own work, but all of them are coordinated with one another, and together they form the whole body.

1. Skeletal system: bones, cartilage and joints form the framework, protect the internal organs, store minerals and produce blood cells.

2. Muscular system: skeletal, smooth and cardiac muscles produce movement; they also help in posture, protection and the movement of substances inside the body.

3. Nervous system: brain, spinal cord and nerves receive stimuli and send impulses; they coordinate the working of all the other systems.

4. Circulatory system: heart, blood and blood vessels transport oxygen, nutrients, hormones and wastes to and from every cell.

5. Respiratory system: nose, trachea and lungs exchange oxygen and carbon dioxide between the blood and the air.
6. Digestive system: mouth, oesophagus, stomach, intestines, liver and pancreas take in food, digest it and absorb the nutrients.
7. Excretory system: kidneys, ureters, bladder and urethra remove the nitrogenous wastes and extra water and salts.
8. Endocrine system: ductless glandular organs (pituitary, thyroid, adrenal and pancreas etc.) secrete hormones which control growth, metabolism and reproduction.
9. Reproductive system: the male and female organs produce gametes and take part in reproduction, so the species continues.
10. Integumentary system: skin, hair and associated structures cover and protect the body, and help in temperature regulation.
11. Immune system: white blood cells and lymphatic organs defend the body against disease-causing organisms.
Coordination: these systems are not separate in working. For example, the muscles need oxygen, so the respiratory, circulatory and nervous systems work with them; the nervous and endocrine systems act as the control centres. Through this coordination the body works as one unit — the level of organization called the organism.

6. Describe the roles of the digestive system and the excretory system in homeostasis.

Digestive system and homeostasis: the digestive system is responsible for the proper distribution of nutrients to cells. Its roles are:
1. it breaks food into nutrients (glucose, amino acids, fatty acids, vitamins, minerals) and absorbs them, so that the blood keeps the correct level of these substances;
2. it reproduces the water and salts that the body needs — the intestine absorbs water, and the body fluids keep their balance;
3. the liver stores extra glucose as glycogen and releases it when the blood glucose falls, so it keeps the blood glucose constant; and
4. it removes undigested waste (faeces) from the body.
Excretory system and homeostasis: the excretory system removes the wastes which would poison the body. Its roles are:
1. the kidneys filter the blood and remove urea, extra salts and excess water as urine, so the composition of the blood stays constant;
2. by controlling the amount of water lost in urine, the kidneys control the water balance (osmotic pressure) of the body;
3. they keep the pH of the blood within the safe range and control the balance of sodium, potassium and other ions; and
4. the lungs remove carbon dioxide and the skin removes some water and salts, so many organs take part in excretion.

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Conclusion: the digestive system supplies and stores the nutrients, and the excretory system removes the wastes. Both work with the nervous and endocrine systems (and with the kidney hormone system) so that the internal environment remains stable, which is exactly what homeostasis means.

7. Explain the functions of various plant organs.

1. Root: it anchors the plant in the soil, absorbs water and mineral salts (with the help of root hairs which increase the surface area), stores food in some plants (carrot, radish), and in legumes it bears nodules with nitrogen-fixing bacteria.

2. Stem: it supports the leaves, flowers and fruits and keeps them in the air for light and pollination; it conducts water and minerals upward (xylem) and food downward (phloem); it stores food in some plants (potato, sugarcane); and in many plants it takes part in vegetative propagation.

3. Leaf: it prepares food by photosynthesis, exchanges gases through stomata, removes extra water by transpiration, and in some plants stores food (aloe) or catches insects (pitcher plant).

4. Flower: it is the reproductive organ; it produces pollen and ovules and takes part in pollination and fertilization, after which it forms the fruit and seed.

5. Fruit and seed: the fruit protects the seed and helps in its dispersal; the seed contains the embryo and the stored food, and germinates into a new plant.

8. Describe the structure and function of the plant root system.

Structure: the root system is the part of the plant below the ground. It develops from the radicle of the seed. There are two types: a tap root system (a main root with lateral roots, as in dicots) and a fibrous root system (many similar roots, as in monocots).

From outside to inside, a root has:

1. Root epidermis (piliferous layer): its cells form outgrowths called root hairs, which increase the surface area for absorption.

2. Cortex: a layer of loosely arranged parenchyma cells that stores food and allows water to pass inward.

3. Endodermis: a single layer with a waterproof band which controls what enters the conducting tissue.

4. Vascular tissue: xylem and phloem arranged in the centre, and in dicots often in a star shape.

Functions:

1. Anchorage: roots fix the plant firmly in the soil.

2. Absorption: root hairs absorb water and mineral salts from the soil.

3. Conduction: xylem carries the absorbed water and salts up to the stem and leaves.

4. Storage: many roots store food, e.g. carrot, radish and sugar beet.

5. Support and other roles: in some plants roots give extra support and help in climbing, and legume roots have nodules of nitrogen-fixing bacteria which enrich the

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soil.

9. Define homeostasis and explain its importance. Discuss how different organ systems work together to maintain homeostasis.

Definition: homeostasis is the ability of an organism to maintain a relatively stable internal environment despite changes in the external environment. It keeps body temperature, pH, water balance and the levels of glucose, salts and gases within a narrow range.

Importance: enzymes work only in a narrow range of temperature and pH; cells need a steady supply of water, glucose and oxygen and must get rid of wastes. Homeostasis therefore keeps metabolism working, protects cells from damage and allows the organism to live in a changing habitat. A failure of homeostasis leads to disease or death.

An example — body temperature: the normal temperature is about 37 °C, controlled by the temperature regulating centre of the brain. When the body becomes cold, the muscles are made to shiver (producing heat) and the skin blood vessels narrow. When the body becomes hot, the sweat glands produce sweat, whose evaporation cools the body, and the skin blood vessels widen. So the same temperature is maintained.

How organ systems work together:

1. Nervous and endocrine systems (control): the hypothalamus, nerves and hormones detect changes and send commands to the other organs — they are the control centre of homeostasis.

2. Respiratory and circulatory systems: the lungs bring oxygen and remove carbon dioxide, and the blood carries these gases to and from the tissues, so the levels of oxygen and carbon dioxide remain balanced; the heart rate and breathing rate change according to need.

3. Digestive system: it absorbs nutrients and water, and the liver stores or releases glucose as needed.

4. Excretory system: the kidneys remove urea and control water and salt balance and the pH of the blood.

5. Integumentary system: the skin protects the body, reduces water loss and helps in temperature control by sweating and by the widening or narrowing of blood vessels.

6. Immune and lymphatic systems: they protect the internal environment from pathogens, so it is not disturbed by infection.

Conclusion: homeostasis is not the work of one organ; it is the result of the coordinated action of all organ systems, controlled by the nervous and endocrine systems. This coordination is the best example of the organism level of organization.

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10. Describe how the respiratory and circulatory systems work together to maintain homeostasis of oxygen and carbon dioxide levels in the body.

1. Oxygen in: during breathing, air enters the lungs and reaches the alveoli, where oxygen diffuses into the blood in the surrounding capillaries because the blood has less oxygen there.

2. Transport: most of this oxygen combines with haemoglobin in the red blood cells, and the heart pumps this oxygenated blood to all the tissues of the body.

3. Oxygen use: in the tissues oxygen leaves the blood and enters the cells, where it is used in cellular respiration to release energy (ATP).

4. Carbon dioxide out: respiration produces carbon dioxide, which diffuses from the cells into the blood. The blood carries it (mainly as bicarbonate) to the lungs, where it diffuses into the alveoli and is removed from the body in the exhaled air.

5. Control by feedback: if the carbon dioxide level in the blood rises, chemoreceptors send impulses to the respiratory centre in the brain, which increases the rate and depth of breathing; the heart rate also increases, so more blood is pumped. If the carbon dioxide level falls, the breathing and heart rates slow down. This is negative feedback.

6. Coordination with other systems: the nervous system controls the rate of breathing and heartbeat, the circulatory system distributes the gases, and the excretory system removes the wastes of respiration. In this way the oxygen and carbon dioxide levels of the body remain within the safe range.

Conclusion: the respiratory system exchanges the gases and the circulatory system transports them, and the feedback control keeps their levels steady — an excellent example of homeostasis by the teamwork of organ systems.

D. Inquisitive Questions

1. How does the structure of epithelial tissue relate to its function in different parts of the body?

Epithelial tissue always covers a surface or lines a cavity, and its form is matched to the work of that particular place:

1. Thin and flat (squamous) epithelium — skin surface, alveoli, blood capillaries: the flat shape makes the surface thin, so gases and other substances can diffuse through it quickly and easily.

2. Cuboidal/columnar epithelium — kidney tubules, intestine: these cells are taller and have more cytoplasm with many mitochondria, which provide energy for absorption and active transport; the columnar lining of the intestine also has microvilli to increase the surface area for absorption.

3. Ciliated epithelium — trachea and bronchi: the cilia beat to sweep mucus and dust out of the air passages, so the structure (hair-like cilia) fits the cleaning job.

4. Glandular epithelium — stomach, sweat glands: these cells are specialized to secrete mucus, enzymes, acid or sweat.

5. Skin (epidermis) with keratin: many layers of flat, dead, keratinized cells make it tough, waterproof and protective, which suits the outer surface of the body.

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6. Root hair cells (plant epidermis): long extensions greatly increase the surface area for absorption of water and minerals.

Conclusion: in every case the shape, number of layers, presence of cilia or microvilli and the thickness of the epithelium decide whether that tissue protects, absorbs, secretes or allows diffusion. Structure and function go together.

2. Evaluate the importance of organ systems working in harmony and predict the consequences of a failure in one system on the others.

Importance of harmony: the organ systems of the body are interdependent. The digestive system supplies glucose, the respiratory system supplies oxygen, the circulatory system carries both to cells, the excretory system removes wastes, and the nervous and endocrine systems control all of them. When they work in harmony, the internal environment stays stable (homeostasis) and the body can grow, act and fight disease. Coordination also makes the body efficient — the heart rate rises exactly when the muscles need more oxygen.

Consequences of failure in one system:

1. Kidney (excretory) failure: urea and extra salts are not removed and water balance is lost. Urea collects in the blood (uraemia) and damages the brain, heart and other organs; the patient needs dialysis or a kidney transplant.

2. Heart (circulatory) failure: oxygen and nutrients do not reach the tissues, so the lungs, kidneys and brain are affected; the person feels weak and breathless, and the kidneys may stop working.

3. Lung disease: less oxygen enters the blood and carbon dioxide collects, which increases the heart rate, disturbs the pH of the blood and weakens all the tissues.

4. Digestive system failure: nutrients are not absorbed, so the cells have less glucose and proteins; the immune system weakens and growth stops.

5. Nervous system failure: because it is the control centre, the messages to the heart, lungs, muscles and glands stop — the coordination of all the systems breaks down.

6. Endocrine (hormone) failure: for example, if insulin is not produced (diabetes), the blood glucose rises and damages the kidneys, eyes, heart and nerves.

Conclusion: the body is a team of systems, so no system can fail alone. A failure in one system quickly spreads as a chain of problems to the others. That is why a healthy life style and timely treatment are important: protecting one system protects the whole body.

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Points to Remember

Levels of organization: organelle → cell → tissue → organ → organ system → organism. A higher level shows **emergent properties** — new capacities that appear only when the parts work together.

Tissue: a group of similar cells that work together to perform a specific function. **Organ:** a structure made of two or more tissues that work together for specific functions. **Organ system:** a group of organs working together for a particular function.

Animal tissues: epithelial (covering, protection, absorption, secretion), connective (joining and support: bone, cartilage, blood), muscular (movement: skeletal, cardiac, smooth) and nervous (impulses).

Plant tissues: epidermal (protective; guard cells and root hairs), ground (palisade and spongy mesophyll) and vascular (xylem and phloem).

Stomach — four tissue layers: epithelium (mucus, HCl, pepsinogen), inner connective tissue (blood vessels and nerves), smooth muscle (churning of food) and outer connective tissue (serosa).

Plant organs: root, stem, leaf, flower. **Root system** (below the ground) absorbs water and minerals and anchors the plant; **shoot system** (above the ground) supports, photosynthesizes, transports and reproduces.

Human organ systems (11): skeletal, muscular, nervous, circulatory, respiratory, digestive, excretory, endocrine, reproductive, integumentary (skin) and immune systems.

Homeostasis is the ability of the organism to maintain a relatively stable internal environment in spite of changes outside. Example: body temperature is kept at about 37 °C by shivering in cold and sweating in heat, under the control of the temperature regulating centre of the brain.

Control of homeostasis: nervous and endocrine systems are the control centres; respiratory and circulatory systems maintain oxygen and carbon dioxide; digestive system supplies nutrients; excretory system removes wastes and controls water and salts; skin helps in temperature control; immune system defends the internal environment.

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Organ system	Main function
Skeletal	Support, protection, storage of minerals, production of blood cells
Muscular	Movement and posture
Nervous	Control and coordination through impulses
Circulatory	Transport of gases, nutrients, hormones and wastes
Respiratory	Exchange of oxygen and carbon dioxide
Digestive	Digestion and absorption of food

Organ system	Main function
Excretory	Removal of nitrogenous wastes; water and salt balance
Endocrine	Secretion of hormones controlling growth, metabolism and reproduction
Reproductive	Production of gametes; continuity of the species
Integumentary	Protection and temperature regulation
Immune	Defence against disease-causing organisms

Table: Human organ systems and their main functions

How to attempt these questions in the paper

• MCQ: write only the correct option (a, b, c or d).

• Short question: 2 to 4 sentences; write the name of the tissue or organ first, then its function.

• Difference question: always draw a two-column table with headings.

• Detailed answer: definition, then numbered points, then a conclusion line. Diagrams of the stomach wall and leaf carry extra marks.

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